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Digital Disruption through Artificial Intelligence in Retail

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Introduction

The swift progress of digital technology and consumers' growing sophistication have made artificial intelligence (AI) the major digital disruptor in the retail sector. In contrast to digitalization at early levels that mainly put the focus on automation and online channel expansion, AI-based technologies intrinsically change how retailers deal with consumer behavior analysis, operation management, and personal shopping experiences through machine learning, predictive analytics, and intelligent automation. In the trade context, AI achieves demand forecasting, inventory optimisation, frequent price adjustment, and customer engagement across both offline and online stores, making it an important element of strategic transformation of an organisation. This paper takes a systematic literature review approach to investigate peer-reviewed studies on theoretically based digital disruption of the retail industry by AI. The review area mainly covers operational, customer experience, and AI adoption, as well as how those changes are explained theoretically. The purpose of this review is to comprehensively synthesize the existing literature, to define such main themes, controversies, and research gaps, and to assess the impacts of AI on the creation of value in this sector and sustainable transformation of the business in the digital economy.

Literature Review Method

A structured literature review method is used in this study to guarantee transparency, precision, and research reproducibility. Through well-known scholarly databases, the peer-reviewed academic articles were collected from them, such as Scopus, Web of Science, ScienceDirect, and Taylor & Francis Online, as well as Google Scholar. These repositories were the ones that were chosen because of their broadness in terms of coverage and upholding of worthy research in business, management, and information systems.

The search process was conducted through the use of keywords that included "artificial intelligence AND retail," "AI-enabled retail," "digital disruption," "AI customer experience," and "AI business transformation." Boolean tools were utilized to bring about the searchers and to keep the search relevant. The first search was filtered using titles and abstracts, which were then followed by a full-text review.

The inclusion criteria specified that the studies be peer-reviewed journal articles that specifically address AI applications in the retail sector, the implications of such applications be strategic, operational, or customer-related, and the publication date from either 2020 to 2025 to highlight the currency. Articles were not accepted if they were purely technical, ripped off their face value of business relevance, and were not explained in a retail context.

Summary Table

<i>Paper Title</i>	<i>Authors & Year</i>	<i>Context / Setting</i>	<i>Key Findings</i>	<i>Short Critical Summary</i>
Application of Artificial Intelligence in Marketing Strategy: A Case Study in the Retail Industry	Wangdra et al. (2025)	Large retail firms; AI-enabled marketing strategy	AI / ML in Personalization, Dynamic Pricing, Demand Forecasting and Building Customer Loyalty	Tangible managerial implications almost entirely descriptive with little empirical validation
Revolutionizing Retail: The Synergy of AI and AR	Gajjar (2024)	Global retail organizations usage of AI & AR for augmented reality	Sales conversion + 35%, returns - 40%, improved efficiency	Perspective integrating different series; based on literature sources and case data primarily
The Mobile Retail Revolution: AI's Transformative Impact on Consumer Behavior and Industry Dynamics	Chanddru (2025)	Mobile commerce & omnichannel retail	Higher repeat purchases, personalization, and inventory accuracy	Strong on tech, weak on consumer behaviour theory
Digital Transformation in SME Retail Operations	Gavhane (2025)	SME liquor retailers in regulated markets (India)	Best quote: "achievement of business value has reduced admin time by percent but improved inventory accuracy by percent."	Rare study of SMEs, sample small means hard to generalise
Artificial Intelligence (AI): Revolutionizing Digital Marketing	van Esch & Black (2021)	B2B and B2C digital marketing	AI improves personalization and marketing efficiency but raises ethical risks	Conceptually strong but lacks primary empirical data
How Artificial Intelligence Will Affect the Future of Retailing	Guha et al. (2021)	Retail senior executives, online and instore retail	AI adds value principally not through customer-facing applications	The AI influence; Predictive, but not empirically tested
The Role of AI in Vendor Performance Evaluation within Digital Retail Supply	Rainy & Chowdhury (2022)	Digital retail supply chains	AI increases the accuracy of vendor evaluation and the ability to detect potential risk	Review; not empirically tested

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Chains				
Changing the Game to Compete: Innovations in the Fashion Retail Industry	Jin & Shin (2020)	Fashion retail	AI demand forecasting mitigates inventory risk	Strong contribution to theory; not to practice
From Data to Decisions: Enhancing Retail with AI and Machine Learning	Dhoni (2024)	Retail organisations using AI and ML systems	Survey AI improves customer insights, inventory optimisation, and decision-making but raises privacy and cost concerns about implementation	Posits strong operational insights but focuses primarily on technical features with limited analysis for how organisations might implement them
AI's Invisible Touch: Customer Experience and Engagement in Online Retail	Lopes et al. (2025)	Survey of AI in retail	AI aims to improve seamless browsing and customer engagement for online retail platform	Strong focus on experiential lead by ethical threat

Findings and Discussions

From the review of the available literature reviewed this research finds three major themes connected to the use of AI within the Retail Sector and how Retail Businesses are going through disruption from the use of AI: Improving Operations, Delivering Superior Customer Experiences, Creating New Business Models. Using AI to Improve Operational Performance.

AI and Operational Transformation

AI's impact on operational effectiveness has been well-documented in numerous articles dealing with the application of artificial intelligence (AI) in retail. Jin & Shin (2020) contend that the integration of artificial intelligence into the supply chain is particularly disruptive in case of fashion retailers where the volatility of demand is considerable, as opposed to traditional demand forecasting approaches used to maintain supply chains. Gavhane (2025) argues that AI-enabled inventory systems not only drastically lessen the amount of time and

resources that retailers need to devote to scheduling and managing their inventories but also greatly improve the precision of their forecasts.

The operational advantages of AI-enabled inventory management are further supported by the results of the Case-Based Research team (2025), who conducted a study involving AI-enabled, real-time marketing strategies (e.g., dynamic pricing, product recommendations, demand forecasting) to help retailers take advantage of changes occurring in their markets while at the same time lowering their operational costs and reducing their risk associated with holding too much inventory. Their study reveals that retailers using AI-enabled Operational Intelligence (AIIO) benefit from higher levels of operational efficiency and can leverage these efficiencies to support pricing and promotional decisions, connecting the optimisation of operation with revenue generation.

However, there are limitations to the existing body of literature regarding the operational improvements provided by AI and other new digital marketing technologies. Most literature is confined to single country and/or industry studies, limiting their applicability to other environments. The majority of literature focusing on operational improvements tends to focus on shorter-term performance measures, therefore the ability of companies to build or develop their capabilities, through the use of artificial intelligence, is largely unexplored.

AI and Customer Experience

There is another area of literature dedicated to using AI as a tool for improving the experiences provided to customers through customer engagement; Lopes et al. (2025) have presented substantial evidence supporting the importance of AI-enabled ease of use to perceived control, concentration, and cognitive enjoyment when utilizing online shopping; the emotion of awe is found to be a strong mediating variable in increasing the likelihood of making a purchase. In addition, Chanddru (2025) and Gajjar (2024) describe how AI can enhance individualization, immersion, and customer loyalty in their respective research.

Wangdra et al. (2025) also confirm that the way AI personalizes through recommendation

systems and targeted promotions significantly contributes to developing a customer's long-term loyalty to a brand or business (as opposed to only making an immediate sale). However, the case studies presented show that customers have high levels of trust in companies that implement proper data-management practices and maintain responsible governance over the use of customer data; therefore, providing customers with enhanced experiences and employing ethical governance of their data are highly interdependent.

While the above research provides valuable insight into how AI can enhance customer experience, the literature does not address potential negative consequences associated with AI implementation, including algorithmic bias, decreased in-person customer interaction, and privacy issues. While these negative impacts are recognized in previous studies, they have yet to be thoroughly investigated from an empirical standpoint, indicating a disconnect between the expectation of technological advancements and the perception of risk by customers.

AI and Business Model Innovation

An additional theme is AI's potential to act as a catalyst for the transformation of existing business models and strategy. By using the example of logistics and supply chain management, Guha et al. (2021) state that AI performs best when used in non-customer-facing operations, rather than as a substitute for frontline workers. In their study, Jin and Shin (2020) refer to how disruptive business models developed by AI technologies can co-exist alongside the traditional retail model.

Building off these ideas, Rainy and Chowdhury (2022) show how AI-based systems of evaluating vendor performance substantially alter how the digital retail supply chain will function. They are able to do this by providing retailers real-time risk detection, predictive analysis, automated decision-making support, and many other things. Their findings indicate that the structure upon which AI applications depend on, such as the quality of the data being sent upstream and the quality of the supplier's knowledge, demonstrates that customer-centric ideals are mistaken when they assert that disruptions of AI will primarily

originate from demand. Though each theme has its own limitations, it is important to note that all four themes are based on cross-sectional and qualitative designs. Researchers do not yet know how AI will develop over time or how factors such as corporate culture, corporate governance, and a company's ability or inability to create a learning environment will affect sustainable results.

Theoretical Understanding

The literature reviewed uses several forms of theory in addition to "The Theory of Disruptive Innovation", which is the basis for studies such as that performed by Jin and Shin (2020) show how AI models lead to the creation of new ways to create value in Retail. Alternatively, the work performed by Rainy and Chowdhury (2022) shows that AI is generally an augmentation of existing systems rather than an absolute substitute and, therefore, retail disruptions may be described as more of an evolutionary change rather than an absolute substitution.

The Technology Acceptance Model (TAM) and Flow Theory serve greater use in consumer-centric studies. The study by Lopes et al. (2025) provides a broader understanding of how AI-enabled engagement has emotional responses (i.e. feelings of awe, etc.) to an end-user while Wangdra et al. (2025) indirectly contest the idea that knowing how to use an AI system (usability) is the only factor in positive consumer reaction; there is evidence that positive consumer reaction is also governed by data governance and ethical trust, neither of which are included in the original idea of the TAM.

Dynamic Capabilities Theory is of particular interest in explaining how AI enables the adaptability of firms. Guha et al. (2021) define AI as a strategic resource, while Rainy and Chowdhury (2022) illustrate that AI allows firms in the Retail Supply Chain to enhance their sense of smell and/or their ability to reconfigure. However, the empirical measurement and operation of Dynamic Capability Theory is still a barrier to providing a greater understanding

of how it can be used in practice.

Overall, current theoretical frameworks provide many useful insights; however, to fully understand how AI affects Retail, through multiple levels, there is no one single theoretical framework that can encompass the breadth of the effect of AI. Multiple frameworks of Dynamic Capability Theory, Governance and Ecosystems will need to be integrated to create better empirical measurement capability.

Conclusions and Recommendations

Based on this systematic literature review, AI is a primary contributor to digital disruption within retail, helping change how retail companies operate, how they interact with customers, and how to improve their business model. The literature suggests AI provides greater accuracy when forecasting demand, provides greater ability to provide personalization, increased price flexibility through AI and an increase in supply chain intelligence (i.e. the ability to make more informed supply chain decisions).

AI transformation of the retail industry will depend on more than just the technological sophistication of AI, but also will depend on adequate data governance, the ability to effectively integrate suppliers, and developing organisational capabilities within organisations. There are several misleading descriptions of AI that suggest AI is solely a customer-facing innovation; this literature review suggests otherwise.

Some of the more significant areas requiring future research include longitudinal study designs, ethical governance of AI, and the ability of SMEs to adopt/utilise AI in their operations. From an organisational/managerial standpoint, retailers need to view AI as a strategic organisational capability and invest in appropriate data infrastructure, create opportunities for cross-functional integration, and develop responsible AI practices. Future research studies should also employ multi-level and longitudinal study designs to improve

understanding of how AI will impact and change retail ecosystems over time.

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